Hierarchical Feature Fusion of Transformer With Patch Dilating for Remote Sensing Scene Classification

Xiaoning Chen, Mingyang Ma, *Student Member, IEEE*, Yong Li, Shaohui Mei, *Senior Member, IEEE*, Zonghao Han, Jian Zhao, and Wei Cheng

***Abstract*— Recently, the Transformer-based technique has emerged as a promising solution for modeling contextual information in remote sensing (RS) scenes and has found widespread applications in RS scene classification. However, how to make full use of intermediate features learned in Transformers is of crucial importance in the RS scene classification tasks. Therefore, this article proposes a hierarchical feature fusion of transformer with patch dilating (HFFT-PD), which aims to capture rich contextual information from hierarchical features to enhance the performance of RS scene classification. Specifically, the HFFT-PD model consists of a hierarchical transformer merging (HTM) block and a lightweight adaptive channel compression (LACC) module, in which the HTM is specially designed for the Transformer architecture to bridge the semantic gaps between features from different hierarchical blocks, and the LACC accounts for the significance of distinct channels in the ultimate classification features. In addition, a brand-new Patch Dilating strategy is uniquely designed for the Transformer paradigm, functioning as a reassembly operator predicated on patch features. Contrasting with conventional upsampling techniques, Patch Dilating facilitates upsampling without requiring supplementary information, while concurrently preserving the semantic content of local spatial structure. Extensive and rigorous experiments conducted on the UC Merced land-use dataset (UCM), aerial image dataset (AID), and NWPU-45 datasets, with training ratios of 80%, 50%, and 20%, respectively, demonstrate that our proposed HFFT-PD outperforms the baseline at least by 0.59%, 0.44%, and 0.99%, respectively, showcasing the significant superiority of our HFFT-PD over contemporary state-of-the-art methodologies.**

***Index Terms*— Feature fusion, remote sensing (RS), scene classification, transformer.**

I. Introduction

HIGH-RESOLUTION satellite images provide enhanced spatial resolution, enabling the capture of finer details and features of terrestrial objects, which contributes to more accurate urban planning [1], [2], disaster monitoring [3], [4], geography exploration [5], environment monitoring [6], and other related studies. Although high-resolution satellite images are predominantly commercial, accessing this data may require payment or be subject to licensing restrictions, thus imposing limitations on its availability. However, due to the remote sensing (RS) images being widely used in various applications, especially in RS image analysis and understanding, RS scene classification has still aroused the enthusiasm of research by experts and researchers in this field.

In RS images, scenes refer to the intelligent identification and classification of land use and land cover (LULC) scenes [7], such as cities, farmland, forests, rivers, and more. The goal of RS scene classification is to categorize the scenes in RS images into distinct and meaningful LULC categories based on the image content. Unlike traditional object classification, RS scene classification focuses more on the overall classification of geographical scenes. This approach aims to identify and differentiate scenes that have similar geographical features and semantic meanings, rather than simply classifying individual object categories. Therefore, in RS image scene classification, the combination of multiple object categories and the context relationship among these objects are important factors for obtaining more comprehensive and accurate scene classification results.

Early traditional methods are generally based on handcrafted features reflecting the low-level feature information, and cannot well express the global semantic information of high-level features. Hence, the classification performance is usually not satisfactory. In recent years, with the development of deep learning in computer vision, convolutional neural networks (CNNs) have achieved promising achievements in the field of RS scene classification [8], [9], [10], [11]. Although the CNN-based methods achieve excellent performance [12], [13], [14], there are still great challenges for RS scene classification, because the local convolutional kernels in CNNs are specialized in capturing local features but suffer in acquiring long-range information in images. To alleviate the limitation, researchers propose some strategies, e.g., multiscale fusion [15], [16] and feature pyramids [17]. While these CNN-based methods have made some improvements, their performance is still suboptimal when solving RS images with complex scenes.

Recently, Transformers have gained extensive attention and success in computer vision. Different from CNNs,

Manuscript received 17 June 2023; revised 13 October 2023; accepted 6 November 2023. Date of publication 9 November 2023; date of current version 22 November 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62201445 and Grant 62171381, in part by the Project funded by the China Postdoctoral Science Foundation under Grant 2022M722550, and in part by the Key Research and Development Program of Shaanxi under Grant 2023-YBGY-258. *(Corresponding author: Yong Li.)*

Xiaoning Chen, Yong Li, Shaohui Mei, Zonghao Han, and Wei Cheng are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: chenxiaoning2018@mail.nwpu.edu.cn; ruikel@nwpu.edu.cn; meish@nwpu.edu.cn; hanzonghao@mail.nwpu.edu.cn; pupil_119@nwpu.edu.cn).

Mingyang Ma is with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi 710049, China (e-mail: mamingyang@mail.xjtu.edu.cn).

Jian Zhao is with the School of Information Science and Technology, Northwest University, Xi'an 710069, China (e-mail: zjctec@nwu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3331880

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

Transformers focuses on long-range contextual information which is essential for a comprehensive understanding of image scenes. It was first proposed for natural language processing (NLP) [18] and was extended to the vision transformer (ViT) [19]. Subsequently, many Transformer-based variants have appeared in image classification tasks [20], [21], [22], [23]. With the development of transformer applications, some researchers bring Transformers into the field of RS scene classification. Bazi et al. [24] first applied the ViT model in RS scene classification and achieved an excellent performance. Zhang et al. [25] proposed the transformers for RS (TRS) model, which combines the Transformer blocks with the CNNs, and obtains superior classification performance. Although the Transformer-based methods demonstrated promising potential for RS scene classification [26], [27], they bring many more parameters and a high calculation cost due to the "columnar" structure of ViT [28]. To reduce the computational cost caused by the same size of each block in the transformer model, some hierarchical Transformers have emerged, such as the Pyramid ViT [28], Swin Transformer [29], and Poolformer [30]. They use a shrinking hierarchical structure to decrease the sequence length of the Transformer, which is similar to the structure of CNNs, thereby substantially diminishing the computational cost. The comparison of the structure of "column" and "pyramid" is shown in Fig. 1.

Despite the hierarchical Transformers being able to achieve excellent classification performance while reducing computing costs, it still faces challenges in handling complex RS scenes that display multiple objects of varying sizes and scales. This is due to the fact that most existing hierarchical Transformer methods only extract contextual information at a single scale and rely solely on high-level features of the last block for classification, disregarding the intermediate features that contain detailed spatial and multiscale information. Therefore, these Transformer models cannot achieve satisfactory results when directly applied to RS prediction tasks that require multilevel local information. Actually, each hierarchy of transformer extracts different characteristic information of images, with the deeper layer of the transformer focusing more on long-term contextual dependencies and the shallower layer focusing more on local details information [31]. For this reason, an effective strategy to cope with the diversity of scales and various types of contents in RS scenes is to merge features of different levels, which can help to capture the necessary local spatial information and multiscale information in RS images and improve the accuracy of RS scene classification.

Thereby, according to the above analysis, this article proposes a hierarchical feature fusion of transformer with patch dilating (HFFT-PD) to fully exploit the different levels of features in the TRS scene classification. Nevertheless, simply feature fusing may cause redundant information, which negatively affects the performance of the model. To address this issue, the HFFT-PD is designed to not only include a hierarchical transformer merging (HTM) block to explore both long-range contextual dependencies and multilevel local characteristics within the scene of RS images but also integrate a lightweight adaptive channel compression (LACC) module

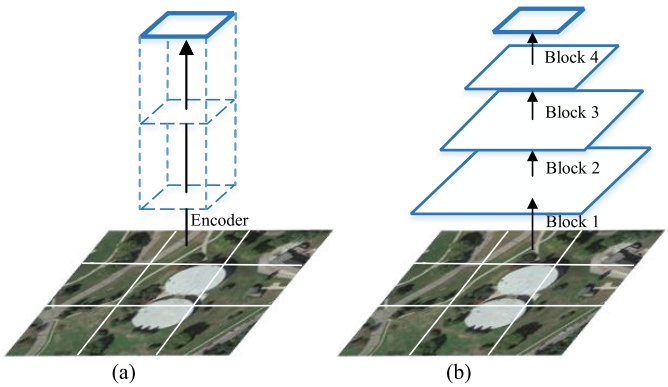


Fig. 1. Architectures of (a) ViT and (b) hierarchical transformer.

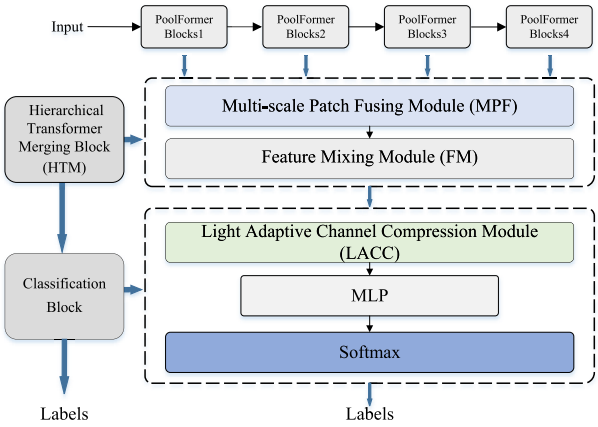


Fig. 2. Flowchart of our proposed HFFT-PD.

to takes into account the significance of the channels to focus on the crucial information and compress the redundant information. The flowchart is shown in Fig. 2.

The HTM block merges the adjacent feature maps to narrow the semantic gaps of different levels and model contextual correlations of cross-level features. However, the conventional future fusion, such as FPN [17], is basically plugged into the traditional CNN backbones but cannot be well applied to Transformer, which processes images based on patches. The proposed HTM is specifically designed for Transformer, which utilizes a novel upsampling strategy named Patch Dilating to upsample the high-level but coarse-resolution feature maps. The Patch Dilating is uniquely designed for the Transformer, which works as a reassembly operator based on the patch. Compared with the existing upsampling methods, Patch Dilating realizes upsampling without introducing any parameters or adding other additional information, while the patches preserve the semantic content of local spatial configurations.

Afterward, to deal with a large amount of information, which not only consumes higher computational and memory costs but also may degrade the performance of the feature representation, the LACC module is proposed to weigh the crucial features and eliminate redundant information to improve the performance of scene classification.

The major contributions of this article are as follows.

1) The HFFT-PD is specially designed for the Transformer architecture and exhibits generality for different hierarchical Transformer models. The HFFT-PD model takes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

into account not only the long-range contextual dependencies within RS images but also the multilevel local characteristics of images.
2) The HTM block is proposed, which consists of a multiscale patch fusing (MPF) module and a feature mixing (FM) module. The MPF merge the adjacent feature maps to narrow the semantic gaps of different level, and the FM module mixes and aggregates these features to model correlations of contextual information of different level features.
3) A novel upsampling method named Patch Dilating is proposed to upsample the high-level but coarse-resolution feature maps, which works as a reassembly operator based on the patch. Meanwhile, several commonly used upsampling methods are compared with ours.
4) A LACC module is designed to weigh the crucial features and eliminate redundant information among different features before the final classifier.

The rest of this article is organized as follows. The related work is introduced in Section II. In Section III, we describe our proposed method in detail. Plenty of experiments are conducted to verify the performance of our proposed method in Section IV. Finally, we conclude this article in Section V.

II. Related Work

A. Transformer

The transformer is first proposed for NLP [18], which relies on encoder-decoder architectures to transform the given sequence. Inspired by the success of NLP, more and more scholars are applying Transformers to other fields, such as object detection [32], image segmentation [33], image classification [34], [35], [36] and other computer vision tasks [37], [38], [39]. Chen et al. [40] proposed iGPT by training the Transformers to auto-regressively predict pixels. However, due to focusing on pixels, these methods consume abundant computational resources and time costs. Different from the previous methods, Dosovitskiy et al. [19] employed the pure Transformer directly to the ViT by feeding the sequences of image patches to this model, which achieves excellent results as well as fewer computation resources are required compared to SOTA CNNs.

ViT just retains the encoder architecture but removes the decoder architecture, and it shows astonishing performance in image classification when pretrained on large-scale datasets [21], [35], [41]. The ViT is a "columnar" structure [28] as shown in Fig. 1(a). For the given input image $\mathbf{X}$, the ViT model first processes it by patch embedding to obtain nonoverlapping patches, each of which is treated as a "token"

$$\mathbf{Z} = \text{PatchEmb}(\mathbf{X}) \tag{1}$$

where $\mathbf{Z} \in \mathbb{R}^{N\times C}$ denote the embedding tokens, the dimension is C and the sequence length is N.

Then, the tokens are fed to the ViT encoder which contains a multihead self-attention (MSA) and a two-layer fully connected feed-forward dense block [multilayer perceptron (MLP)]. The most important subblock is the MSA block which can map the relationship of feature representation among tokens. The MSA comprises four layers, namely, the linear layer, the self-attention layer, the concatenation layer, and the final linear layer. To obtain the query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) matrices for the single-head self-attention layer, the sequence $\mathbf{Z}$ is projected onto three trainable matrices. The calculation is as follows:

$$\begin{aligned}\mathbf{Q} &= \mathbf{Z}\times\mathbf{W}_Q^T\\ \mathbf{K} &= \mathbf{Z}\times\mathbf{W}_K^T\\ \mathbf{V} &= \mathbf{Z}\times\mathbf{W}_V^T\end{aligned} \tag{2}$$

where $\mathbf{W}_Q^T$, $\mathbf{W}_K^T$, and $\mathbf{W}_V^T$ are three trainable matrices. The dot-product is utilized to match $\mathbf{Q}$ and $\mathbf{K}$ to measure the similarity, and then, is divided by $\sqrt{d}$. Then, the Softmax function is used to attain the attention matrix as follows:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}. \tag{3}$$

For the MSA, the input sequence is divided into m groups first. Then, according to (3), each group Attention_i, $(i = 1,\ldots,m)$ is calculated. Therefore, the MSA can be obtained by the concatenate operation and projected through a feed-forward layer with learnable weights $\mathbf{W}^O$ to the desired dimension

$$\text{MSA}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Concat}(\text{Attention}_1,\ldots,\text{Attention}_m)\mathbf{W}^O. \tag{4}$$

Since the ViT block contains a MSA subblock and a two-layer MLP subblock, the ViT block can be expressed as follows:

$$\mathbf{Y} = \text{MSA}(\text{LN}(\mathbf{Z})) + \mathbf{Z} \tag{5}$$

$$\mathbf{T} = \text{MLP}(\text{LN}(\mathbf{Y})) + \mathbf{Y} \tag{6}$$

where LN denotes the layer normalization.

B. Hierarchical Transformer

Although the ViT model achieves excellent performance in scene classification, it still shows some drawbacks: 1) ViT is a "columnar" structure, which makes its output feature map low-resolution and single-scale and 2) the scale of each transformer block is the same, resulting in relatively high calculation cost [28]. To solve the above-mentioned problems, some hierarchical Transformers have emerged that not only generate pyramid feature maps but also reduce the computation cost. The Pyramid ViT [28] is the first to apply the pyramid structure to the Transformer, which takes advantage of a progressive shrinking pyramid and spatial-reduction attention to building a hierarchical Transformer. The Swin Transformer [29] utilizes patch merging in deeper layers to reduce the size of feature maps to build a hierarchical Transformer. Poolformer [30] uses the pooling operation to drop down the scale of feature maps to establish a hierarchical Transformer that replaces the self-attention module with a nonparametric pooling operation. The comparison of the structure of "columnar" and "hierarchical" is shown in Fig. 1. It demonstrates that the approaches achieve excellent results, especially in the image classification tasks.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

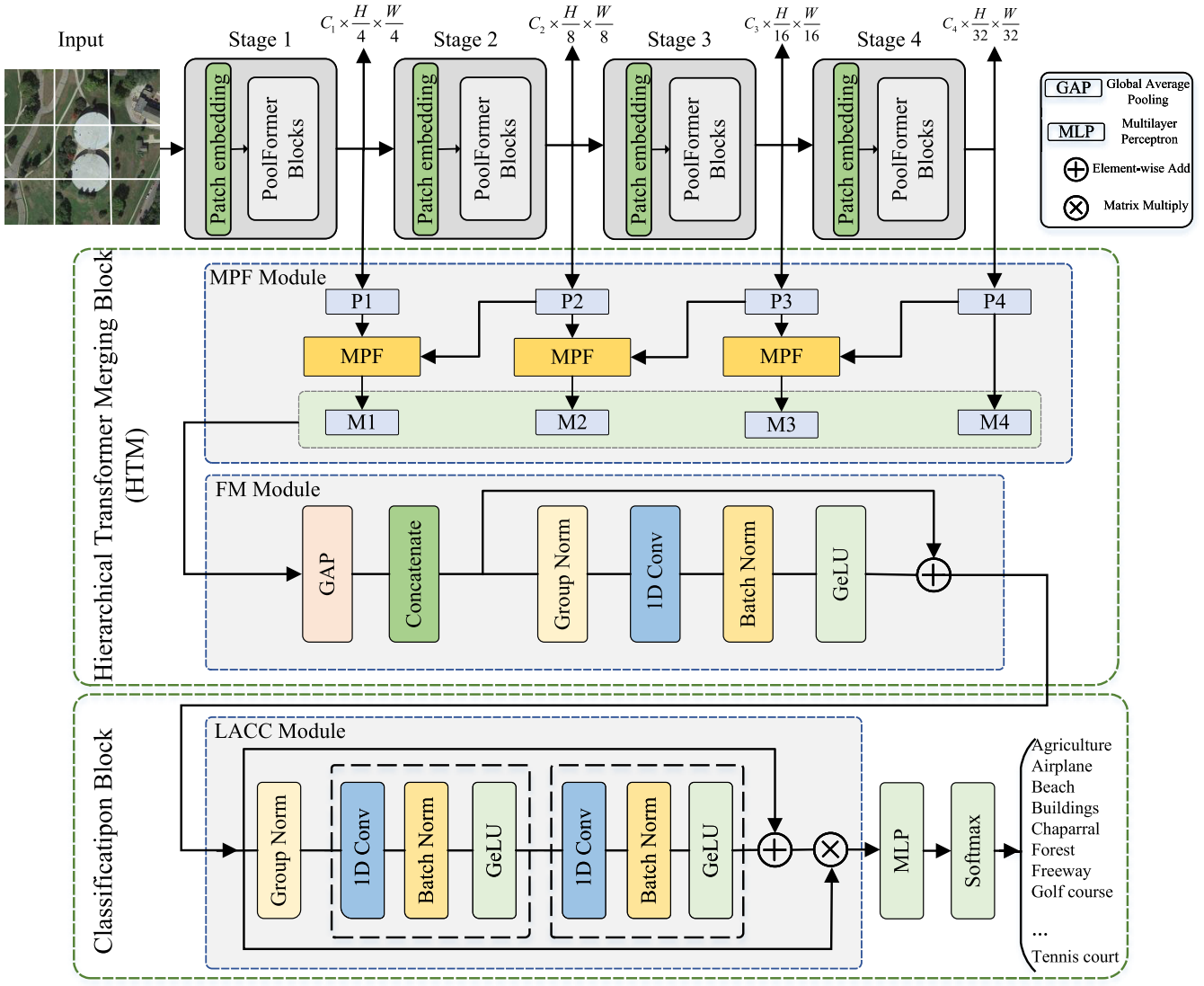


Fig. 3. Architecture of our proposed HFFT-PD. The MPF represents the multiscale patch fusing, the FM module stands for feature mixing, and the LACC stands for the lightweight adaptive channel compression module.

In conclusion, the hierarchical Transformers employs progressively shrinking pyramids, similar to traditional CNNs [42], to reduce the size of large feature maps. This approach enables the hierarchical Transformer to obtain multiscale and high-resolution feature maps in a flexible manner [30].

C. Transformer in RS Scene Classification

With the extensive application of Transformer, some scholars have applied it to the scene classification of RS images [43], [44]. Bazi et al. [24] first introduced the ViT model to RS scene classification which directly applies the original Vit model to RS images, and achieves excellent performance. Lv et al. [27] proposed SCViT which considers not only the geometric information but also channel information for the scene classification of the complex RS scenes. In order to solve the issue that Vit cannot generalize well when faced with insufficient training data, a novel ViT-CL [45] was proposed, which combines supervised contrastive learning with the Vit and achieves superior results. The transformer-based methods demonstrate promising potential for RS scene classification [26].

In addition, since the benefits of convolution and transformer structures are complementary, several researchers investigated integrating these two network architectures. Deng et al. [46] proposed the CTNet which applies a joint loss function to combine CNNs and Transformer to further improve the feature expression ability. To consider the diversity in the type of RS scene contents, Tang et al. [47] proposed a novel EMTCAL that combines the transformer and CNNs to explore the information within RS scenes fully. In order to learn the overall features, Zhang et al. [25] proposed the architecture of "pure CNNs → Convolution + Transformer → pure Transformers" for RS scene classification, and achieved excellent performance. Xu et al. [43] proposed a hybrid network, which applies Vit as a teacher for guiding the CNNs to realize the advantages of both Transformer and CNNs. A lot of work emerged to further improve the performance of the Transformer [48].

Although these methods achieved excellent performance, they ignore the importance of multiscale and high-resolution features which is crucial for the scene classification of RS images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

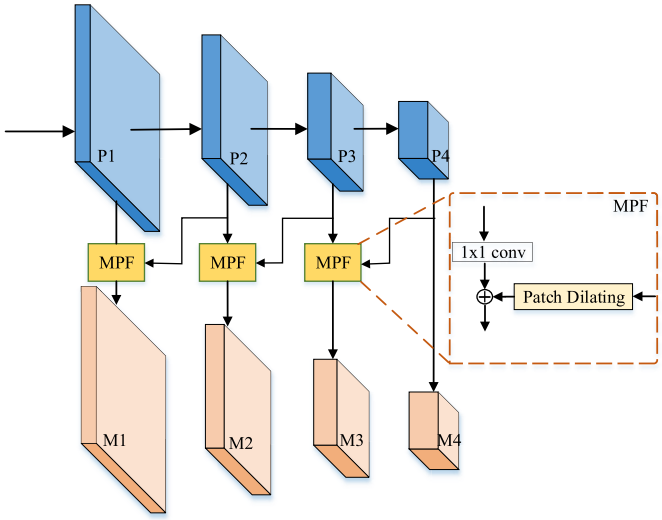


Fig. 4. MPF module.

III. Proposed Method

The architecture of the proposed method is displayed in Fig. 3 (Poolformer is chosen as an instance) which contains three main parts: 1) multiscale and multiresolution features generation; 2) multiscale and multiresolution feature merging; and 3) a classification layer. In the first part, the hierarchical Transformer (e.g., Poolformer, Swin Transformer) is chosen to generate the multiscale and multiresolution features from RS images. The output of the Transformer block in each stage from up to the bottom is the feature candidate. In the second part, we design a HTM block that consists of a MPF module and a FM module. The MPF module merges the adjacent feature maps to narrow the semantic gaps of different levels. This module adopts Patch Dilating upsampling, which works as a reassembly operator based on the patch, to align the spatial resolution of adjacent levels. The details of the HTM block are provided in Section III-A. The third part is the classification block which adds a LACC module to weigh the channel of features. Details of the classification block are introduced in Section III-B.

A. HTM Block

Since the hierarchical Transformer can generate multilevel feature maps, fusing these feature maps to obtain more discriminative features is conducive to RS scene classification. The HTM block comprises two modules: the first is the MPF module, which merges adjacent feature maps, and the second is the FM module, which mixes and aggregates these features.

1) MPF Module: In this part, the MPF module is introduced to learn multiscale and high-resolution features. Firstly, the feature maps are obtained from the last output of the Transformer block in each stage and denoted as $\{P_1, P_2, P_3, P_4\}$. However, there is a big semantic span between different level feature maps, so fusing these features directly is not an effective strategy. To address this issue, the MPF module is proposed to narrow the semantic gap. The details of the MPF module are as Fig. 4.

To align the different sizes of feature maps between adjacent blocks in a hierarchical Transformer, the Patch Dilating operation is first executed, which enlarges the number of tokens by a multiple of $2 \times 2 = 4$ ($2\times$ upsampling of resolution).

Patch Dilating works as a reassembling operator that aligns the spatial resolution of adjacent levels based on the patch, which distinguishes it from traditional upsampling methods used to enlarge the scale of the feature map. Traditional upsampling methods (e.g., Nearest Neighbors or Bilinear Interpolation) not only increase extra redundant information, but also produce discontinuous phenomenon and the edge of the object fuzzy, which makes it difficult to capture rich semantic information [10], and may significantly affect the generation of feature maps of complex RS scenes. Another typical upsampling method is Deconvolution, which brings a large number of parameters to realize adaptive interpolation and thus increases the computation cost [49]. Since the Transformer performs feature transformation based on patches, Patch Dilating can effectively preserve the internal characteristics in patches. It expands the patch resolution of the feature map by transferring the patches in the depth of the channel to the width and height of spatial dimension, which achieves upsampling without introducing any parameters or adding other additional information, and the patches retain the semantic information of the local region, which is beneficial for scene classification. The Patch Dilating is shown in Fig. 5. Note that a color block represents an image patch with the resolution of $r \times r$. The parameter r is the number of the image tokens.

Specifically, the feature map is from one of the feature sets $\{P_i, i = 2, 3, 4\}$, and the size of P_i is assumed to be $C_i \times H \times W$. The feature map of P_i is firstly split into $2 \times 2 = 4$ groups along the channel dimension, as shown in the left of Fig. 5. As a result, the channel dimension will be reduced by a factor of $1/4$. Then, the maps with the size of $1/4C_i \times 2H \times 2W$ are obtained by shuffling these patches regularly. Finally, these features are mapped by convolution operation to align the channel dimension with the adjacent lower layer, denoted as $C_{i-1} \times 2H \times 2W$.

Then these enlarged feature maps are merged with their lower-level feature maps by element-based addition as follows:

$$P_i' = \begin{cases} P_i \oplus \mathrm{PD}(P_{i+1}), & i \in \{1, 2, 3\} \\ P_i, & i \in \{4\} \end{cases} \tag{7}$$

where PD denoted the Patch Dilating operation, and $P_i', i \in \{1, 2, 3\}$ are the merged feature maps after the element-base addition. Then, a 3×3 convolution on each merged map P_i' is applied to obtain the final feature maps, aiming to reduce the aliasing effect of upsampling. The final feature maps are labeled as $\{M_1, M_2, M_3, M_4\}$, corresponding to $\{P_1, P_2, P_3, P_4\}$. Finally, these patches will be fed into the FM module for further feature mixing to obtain more discriminating features.

2) FM Module: The multiscale feature maps are denoted as $\{M_1, M_2, M_3, M_4\}$ which are acquired by the MPF module. Due to the fusion of adjacent features in these feature maps, directly concatenating them can result in an aliasing effect due to their consistent regularity, which diminishes the capability

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

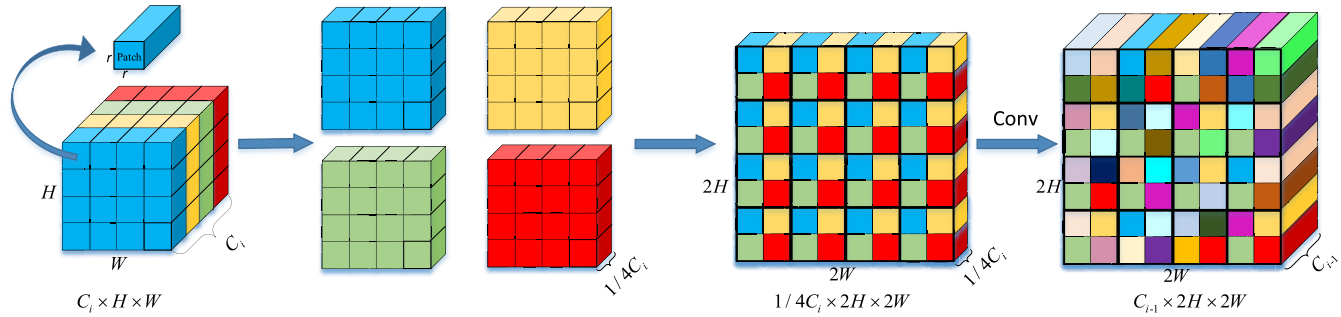


Fig. 5. Details of patch dilating [a color block represents a patch whose resolution is (r, r)]. The patch dilating expands the patch resolution of the feature map by transferring the patches in the depth of the channel to the width and height of spatial dimension.

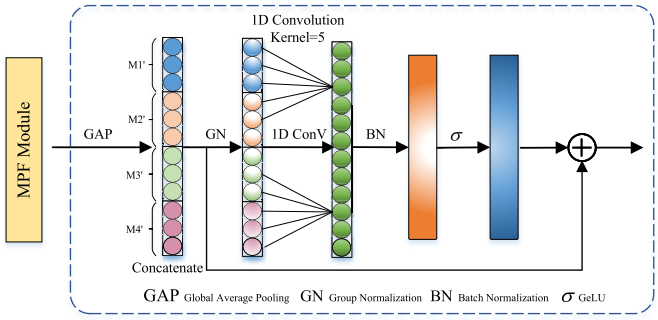


Fig. 6. FM module.

of feature representation. In addition, 1-D convolution can help maintain model lightweightness. Therefore, the FM module is introduced to mix and aggregate these concatenated features and extract more discriminate features to explore correlations of multilevel features. The details of the FM module are displayed in Fig. 6. First, the global average pooling (GAP) operation is applied along the spatial dimensions to aggregate the spatial information of M_i, $i = \{1, 2, 3, 4\}$, and obtain the features labeled as M_i', then these features are concatenated by channel dimension to acquire the fusing features $\mathbf{M} \in \mathbb{R}^{1\times C}$. These features are then aggregated by feeding them into 1-D convolution, to further improve the performance and robustness. Group Normalization and Batch Normalization are added before and after the convolution operation, respectively, to acquire the features $\mathbf{R} \in \mathbb{R}^{1\times C}$. The equation is shown in the following:

$$\mathbf{R} = \sigma\left(\mathrm{BN}\left(\mathrm{ConV1D}_k\left(\mathrm{GN}(\mathbf{M})\right)\right)\right) \oplus \mathbf{M} \tag{8}$$

where $\mathrm{ConV1D}_k$ denotes the 1-D convolutional operation with the kernel size $k = 5$ which is the result of experimental verification, the GN and the BN indicate Group Normalization and Batch Normalization, σ denotes the GeLU function, and $\oplus$ stands for element-based addition.

B. Classification Block

Concatenating multilevel features will cause information redundancy, and affect the classification performance [48]. Therefore, a LACC module is introduced to adaptively weigh these features before classification. Finally, the compressed features are classified by MLP, and a softmax classification layer is followed to obtain the labels.

Extracting key features and removing redundant information is crucial for classification. Therefore, to achieve this purpose, the LACC module is applied before the final classification, which weights the crucial features and eliminates redundant information to enhance the feature representation.

1) LACC Module: To achieve excellent performance with fewer parameters, our LACC module adopts a two-layer 1-D convolution operation to compress and highlight the classification features. As shown in Fig. 7.

First, the Group Normalization is used to normalize the features from the HTM block, the two-layer 1-D convolution is successively adopted, while a Batch Normalization and a GeLU function are added after the convolution operation, respectively. The equation is shown in the following:

$$\mathbf{F}' = \sigma\left(\mathrm{ConV1D}_m\left(\sigma\left(\mathrm{ConV1D}_n(\mathbf{R})\right)\right)\right) \tag{9}$$

where $\mathrm{ConV1D}_m$ and $\mathrm{ConV1D}_n$ denote the 1-D convolutional operation with the kernel size $m = 5$ and $n = 5$, respectively, Group Normalization and Batch Normalization are added before and after the convolution operation, and σ denotes the GeLU function. Subsequently, a residual connection is obtained to highlight and compress the classification features, as the following:

$$\mathbf{F} = (\mathbf{F}' \oplus \mathbf{R}) \otimes \mathbf{R} \tag{10}$$

where "$\otimes$" represents the element-wise multiplication operation, while "$\oplus$" denote to the element-wise addition operation.

2) Classification Layer: After attaining the weighted feature vectors, a MLP and softmax layer are utilized to acquire the scene label of the input RS.

Given $f_i, (i = 1, 2, \ldots, n)$ is the output of the MLP, where n is the total number of the scene labels. The formula for the Softmax function is as follows:

$$\alpha_i = \frac{\exp(f_i)}{\sum_j^n \exp(f_j)} \tag{11}$$

$$\alpha = \max(\alpha_1, \alpha_2, \ldots, \alpha_n) \tag{12}$$

where α_i represent the pseudo-probabilities that the input image attributed to the ith category. The final label is determined by (12).

The cross-entropy loss function is utilized [50] to optimize our proposed classification model. The formula for the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

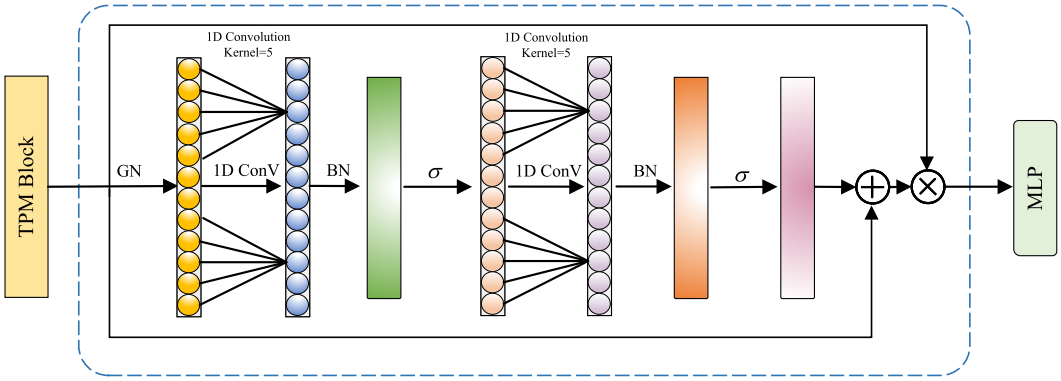


Fig. 7. Details of our proposed LACC module.

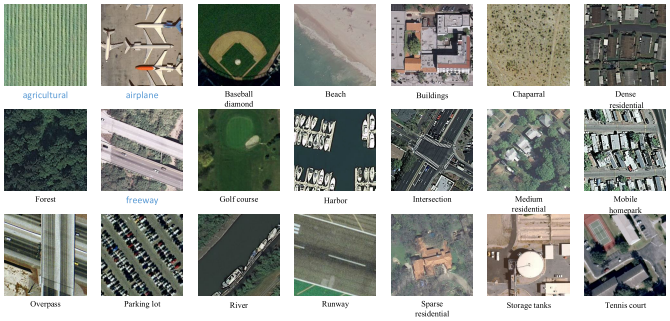


Fig. 8. Example images of UCM dataset.

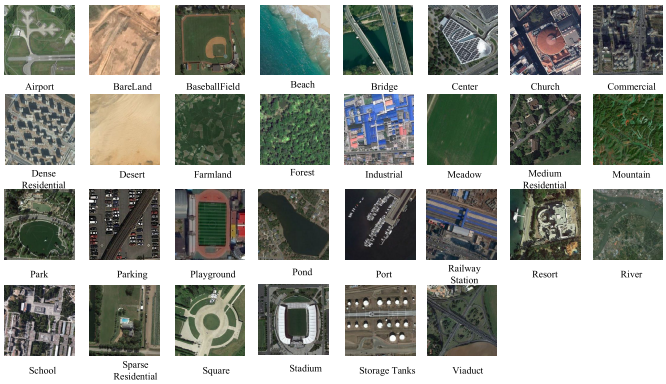


Fig. 9. Example images of AID dataset.



Fig. 10. Example images of NWPU dataset.

cross-entropy loss function is given below

$$\text{Loss} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C} y_c^n \log\left(\hat{y}_c^n\right) \tag{13}$$

where y stand for the real scene label, $\hat{y}$ represent the predicted scene label, C is the number of categories, and N is the number of samples in a mini-batch.

IV. Experiments and Analysis

A. Dataset

To demonstrate the performance of our proposed HFFT-PD, three popular image benchmarks are employed in experiments including the UC Merced land-use dataset (UCM) [7], the aerial image dataset (AID) [51], the NWPU-RESISC45 dataset (NWPU) [7]. The three benchmark datasets pose unique challenges, as follows.

1) These three datasets stand for small, medium, and large datasets that enable our method to be evaluated more comprehensively.
2) The RS images show complex spectral and spatial distributions because of the different sensors.
3) There are great differences in the categories of these three datasets, which are more difficult for scene classification.

The details of the three datasets are as follows.

1) UCM: The UCM dataset, which contains 2100 RS scene images and 21 different classes, was collected by the University of California at Merced. The images, 256 × 256 pixels in size, offer a spatial resolution of 30 cm/pixel and come from a large range of locations across the United States. The details of the UCM dataset are shown in Table I, and one image is selected from each category in the UCM dataset, which is shown in Fig. 8.

2) AID: The AID dataset is a multisource dataset with more categories and a larger scale compared to the UCM dataset and therefore has a greater challenge. The AID dataset was collected by Wuhan University, which covers different areas of different countries. This dataset contains a total of 10 000 RS images with 30 classes. The size of each image is 600 × 600 with ranges from about 0.5 to 8 m/pixel of spatial resolution. The details of the AID dataset are shown in Table I,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED DESCRIPTION OF EACH DATASET

Dataset	Size	Spatial resolution	Number of classes	Total number
UCM	256×256	30cm	21	2100
AID	600×600	0.5m-8m	30	10000
NWPU	256×256	0.2m-30m	45	31500

TABLE II
DATA SETTING IN THE EXPERIMENTS

Datasets	UCM	AID	NWPU
Split-1	80% training 20% testing	50% training 50% testing	20% training 80% testing
Split-2	50% training 50% testing	20% training 80% testing	10% training 90% testing

and one image is randomly selected from each category in the AID dataset as shown in Fig. 9.

3) NWPU: The NWPU dataset, collected by Northwestern Polytechnic University, is a large-scale RS dataset consisting of 31 500 RS images with 45 classes. Each image is 256×256 pixels in size, with varying spatial resolutions ranging from 0.2 to 30 m/pixel. The details of the NWPU dataset are shown in Table I, and one image is randomly selected from each category in the NWPU dataset as shown in Fig. 10.

B. Experimental Setup

1) Data Settings: In the experimental dataset, each dataset is randomly divided into two parts. In all subsequent experiments, the training set and the testing set are completely independent datasets, and the detailed data splitting is shown in Table II. Two splits of the data were carried out for each dataset, as shown in Table II. To further improve the generalization performance of the models and prohibit overfitting in the training stage, data augmentation including random flipping vertically and horizontally, illumination adjustment, random rotation, and so on, is employed.

2) Implementation Details: To expedite the computations, an NVIDIA RTX 3080 was utilized, and all the experiments were carried out using the PyTorch library. The model of the backbone is first pretrained on the ImageNet dataset and then fine-tuned on the RS datasets with 200 learning epochs, and the batch size is set as 32. Cosine learning and the AdamW optimizer [58] is employed as the training strategy, and the initial learning rate is set to $1e^{-4}$. All the RS images are resized into 224×224. The size of patches is set to $r = 2$. The experiments are repeated five times with the corresponding means and standard deviations.

3) Evaluation Metrics: In this article, the performance of the method is quantitatively evaluated using the overall accuracy (OA) and confusion matrix (CM) as selected metrics. The OA indicates the number of correctly classified images divided by the total number of images. The CM is used to represent the probability that each category is predicted to belong to the other categories. In addition, the $F1$-score and Specificity are obtained to further evaluate the performance of the proposed model.

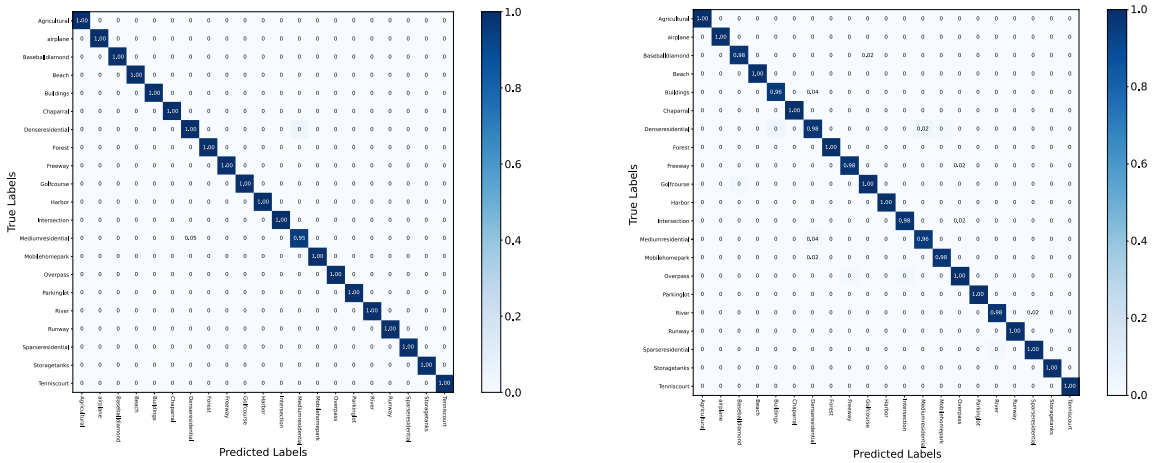
TABLE III
COMPARISON OF THE OA (%) ON THE UCM DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Overall Accuracy(%)	
	80%	**50%**
CaffeNet [7]	95.02±0.81	93.98±0.67
GoogLeNet [7]	94.31±0.89	92.70±0.60
VGG-VD-16 [51]	95.21±1.20	94.14±0.60
ResNet50 [52]	96.98±0.49	95.69±0.23
TEXNet [53]	96.79±0.49	95.89±0.37
APDC-Net [54]	97.05±0.43	95.01±0.43
VGG-VD-16-CapsNet [55]	98.81±0.22	95.33±0.18
ResNet50-EAM [52]	98.98±0.37	/
LCNN-BFF [56]	99.29±0.24	94.64±0.21
EFPN-DSE [10]	98.12±0.57	96.65±0.36
LG-CNN [13]	98.61±0.22	98.97±0.31
DFAGCN [57]	/	98.48±0.42
Vit-B [19]	99.01±0.21	98.48±0.39
PVT [28]	98.81±0.36	98.38±0.39
Poolformer [30]	99.02±0.28	98.29±0.41
Swin [29]	99.29±0.31	98.76±0.27
T2T-ViT-12 [21]	99.10±0.32	97.79±0.20
SCViT [27]	99.52±0.31	98.90±0.19
METCAL [47]	99.57±0.28	98.67±0.16
HFFT-PD-PVT (ours)	99.82±0.31	**99.33±0.32**
HFFT-PD-Poolformer(ours)	99.86±0.27	99.12±0.31
HFFT-PD-Swin(ours)	**99.88±0.32**	99.16±0.41

C. Comparison Experiments

1) Comparison of OA With State-of-the-Arts: To comprehensively verify the superiority of our proposed algorithm, both CNN-based algorithms and Transformer-based algorithms are compared, respectively. For two splits of the same dataset in Table II, the proposed method is restarted for training and testing which are two separate experiments, and in all subsequent experiments, the testing dataset will be used for model evaluation. CaffeNet [7], GoogleNet [7], VGG-VD-16 [51], and ResNet50 [52], are four well-known and commonly used CNN-backbone models. TEXNet [53], ARCNet [59], APDC-Net [54], VGG16-CapsNet [55], VGG-VD-16-SAFF [60], LCNN-BFF [56], DFAGCN [57], are for some state-of-the-art CNN-based methods. For the Transformer-based models, Vit [19], PVT [28], Swin [29], Poolformer [30], SCViT [27], METCAL [47] and MSFT [48] are selected for comparison. If the code can be obtained, those compared methods are trained and tested with the default settings, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 11. CMs of HFFT-PD on the UCM dataset with (a) 80% and (b) 50% training ratios.

while the corresponding code is not accessible, we relied upon the original results presented in the authors' works. In order to balance the number of model parameters and classification accuracy, Vit-base is used for the ViT model, Swin-tiny for the Swin model, and Poolformer-base for the Poolformer as the backbone of the proposed models in the following experiments.

a) Experimental results on UCM dataset: The contrastive experiments are conducted on the UCM dataset, which was randomly split into 80% and 20% as training datasets and testing datasets, as well as 50% and 50% as training datasets and testing datasets, and the results are shown in Table III. The best results are highlighted in bold. As can be found from Table III, the CNN-based approaches have higher classification results than the classical CNN models, while the Transformer-based approach achieves significantly higher accuracies than the CNN-based approaches. In the CNN-based methods, the LCNN-BFF performs best, obtaining a classification OA of 99.29% at the 80% training ratio. The worst result of our proposed method is 99.82% produced by HFFT-PD-PVT and is still higher than the LCNN-BFF method for 0.53%. In the same way, our method is 0.15% higher than the best CNN-based method under the 50% training ratio. Compared to the Transformer-based methods using the same way as above, our method outperforms the Transformer-based method by 0.25% and 0.22% at the 80% and 50% training ratio, respectively. This is because the aggregation strategy of the CNN-based approach produces more discriminative features than the CNN backbone, while the transformer-based approach exploits global contextual information, and the proposed model not only takes advantage of Transformer but also considers multiscale feature information. These results illustrate that our method has superior classification performance on the UCM dataset.

To further illustrate the advantage of our method, the CMs of HFFT-PD-Poolformer are also reported in Fig. 11. As can be seen from Fig. 11 most of the categories are correctly predicted. From Fig. 11(a), the confusion only occurs in "Mediumresidentia" and "Denseresidential" under the 80% training ratio. When the training ratio is reduced to 50%, there are still 13 of 21 categories are correctly predicted as can be seen in Fig. 11(b), and 19 of 21 categories achieve an accuracy of over 98%. The above discussions indicate that the proposed HFFT-PD is beneficial for the UCM dataset.

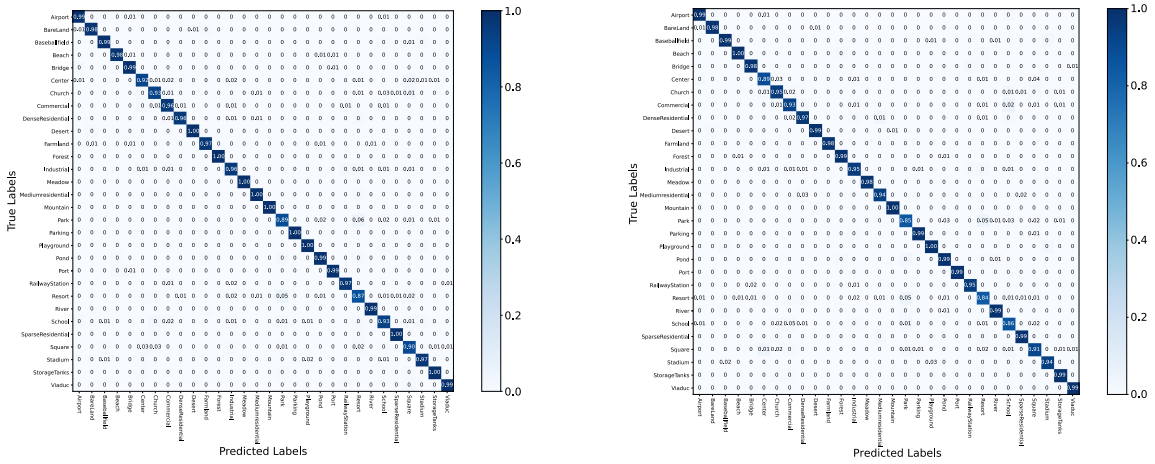
TABLE IV

COMPARISON OF THE OA (%) ON THE AID DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Overall Accuracy(%)	
	50%	20%
CaffeNet [7]	89.53±0.31	86.86±0.47
GoogLeNet [7]	86.39±0.55	83.44±0.40
VGG-VD-16 [51]	89.64±0.36	86.59±0.29
ResNet50 [52]	95.96±0.13	93.64±0.25
TEXNet [53]	92.96±0.18	90.87±0.11
APDC-Net [54]	92.15±0.29	88.56±0.29
VGG-VD-16-CapsNet [55]	94.74±0.17	91.63±0.19
ResNet50-EAM [52]	96.62±0.13	93.63±0.25
LCNN-BFF [56]	94.62±0.16	91.66±0.48
EFPN-DSE [10]	93.35±0.09	89.28±0.05
LG-CNN [13]	97.65±0.25	94.74±0.23
DFAGCN [57]	94.88±0.22	/
Vit-B [19]	96.49±0.18	94.90±0.29
PVT [28]	97.07±0.26	95.59±0.16
Poolformer [30]	96.41±0.18	94.32±0.26
Swin [29]	96.75±0.32	96.24±0.36
T2T-ViT-12 [21]	96.29±0.24	94.39±0.22
SCViT [27]	96.98±0.16	95.56±0.17
METCAL [47]	96.41±0.23	94.69±0.14
HFFT-PD-PVT (ours)	97.51±0.23	96.04±0.28
HFFT-PD-Poolformer(ours)	97.34±0.32	96.01±0.37
HFFT-PD-Swin(ours)	**97.91±0.27**	**97.08±0.53**

b) Experimental results on AID dataset: For further illustration of the superiority of the proposed approach, some state-of-the-art methods are compared on the AID dataset. In the experiment, the AID dataset was randomly split into 50% and 50% as training datasets and testing datasets, as well as 20% and 80% as training datasets and testing datasets. The results for the AID dataset are shown in Table IV. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 12. CMs of HFFT-PD on the AID dataset with (a) 50% and (b) 20% training ratios.

best results are highlighted in bold. Similar conclusions are obtained on the AID dataset as the UCM dataset. Specifically, the LG-CNN [13] achieve the best results of 97.65% and 94.74% at the 50% and 20% training ratio, respectively, in the CNN-based methods, while the proposed method of HFFT-PD-Swin is higher than LG-CNN 0.26% and 2.34% at the 50% and 20% training ratio, respectively, which is because our approach not only considers the long-distance context dependence within RS scene but also takes into account multilayer feature fusion. Although the HFFT-PD-PVT and HFFT-PD-Poolformer have slightly lower classification results than LG-CNN at 50% of the training samples, they are much higher than LG-CNN for 20% of the training samples. For Transformer-based methods, our method still performs relatively well, especially HFFT-PD-Swin is higher than SCViT [27] 0.93% and 1.52% at the 50% and 20% training ratio, respectively. This implies that the contextual relationship excavating mechanism and multiscale feature learning strategy can adequately capture the sufficient and diverse land cover in RS scenes. The above constructive results validate the effectiveness of our method.

Besides, the CMs of our HFFT-PD-Poolformer are shown in Fig. 12. Take Fig. 12(b) as an example, when the training ratio is 20%, it is evident that 90% accuracy was achieved in 26 out of 30 categories, while only the "Center," "Park," "Resort," and "School" have classification accuracies under 90%. Even though the "Resort" has high interclass similarity with the "Park," only 5% is wrongly classified into the category of "park." Based on the results above, it is evident that our method maintains excellent classification performance even training ratio is reduced to 20%.

c) Experimental results on NWPU dataset: For the NWPU dataset, it was randomly split into 20% and 80% as training datasets and testing datasets, as well as 10% and 90% as training datasets and testing datasets. Table V shows the classification results of different methods on the NWPU dataset, in which our proposed HFFT-PD-Swin also reaches the best performance in all cases. Compared with the Transformer-based methods, the enhancements of our proposed HFFT-PD-Swin are 2.79% (over T2T-ViT-12), 1.32% (over SCViT), and 2.33% (over METCAL) under the 20% training ratio. While under the 10% training ratio, the proposed HFFT-PD-Swin is higher 3.36%, 1.26%, and 2.35% than the above methods, respectively. It can also be found that our proposed model with the different backbones of the Transformer obtained a higher OA than the corresponding backbone. It can be seen that our method obtains the best classification results when using Swin as the backbone, no matter the UCM dataset, the AID dataset, or the NWPU dataset.

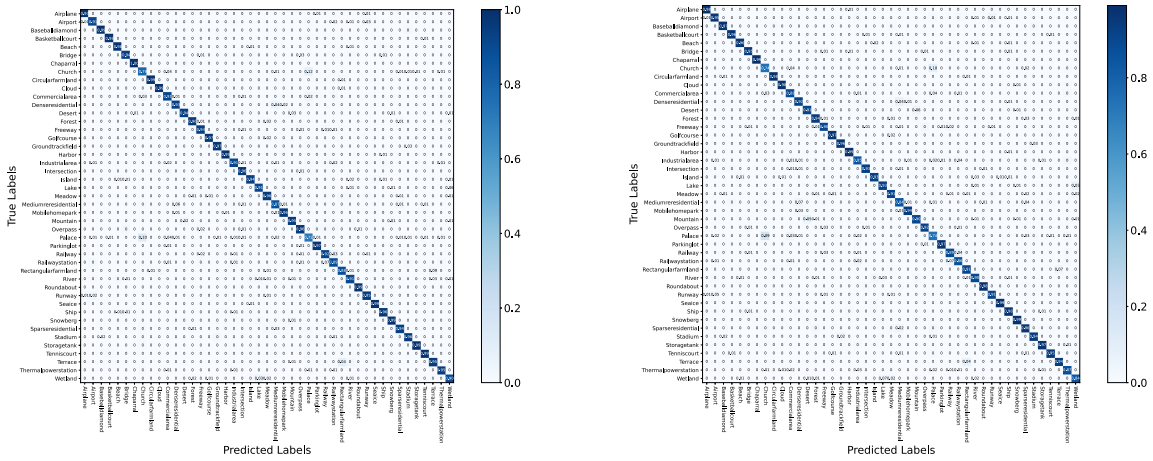
TABLE V

COMPARISON OF THE OA (%) ON THE NWPU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Overall Accuracy(%)	
	20%	10%
GoogLeNet [7]	86.43±0.27	83.44±0.40
VGG-VD-16 [51]	79.79±0.15	76.47±0.18
ResNet50 [52]	91.86±0.19	88.48±0.21
APDC-Net [54]	87.84±0.26	85.94±0.22
VGG-VD-16-CapsNet [55]	89.18±0.14	85.08±0.13
ResNet50-EAM [52]	93.51±0.12	90.87±0.15
LCNN-BFF [56]	91.73±0.17	86.53±0.51
LG-CNN [13]	94.43±0.16	91.91±0.15
DFAGCN [57]	89.29±0.28	/
Vit-B [19]	93.90±0.20	91.59±0.19
PVT [28]	93.77±0.29	92.10±0.35
Poolformer [30]	93.59±0.26	91.38±0.22
Swin [29]	94.89±0.43	93.08±0.26
T2T-ViT-12 [21]	93.19±0.10	90.62±0.18
SCViT [27]	94.66±0.10	92.72±0.04
METCAL [47]	93.65±0.12	91.63±0.19
HFFT-PD-PVT (ours)	94.76±0.51	92.79±0.35
HFFT-PD-Poolformer(ours)	94.63±0.36	92.83±0.32
HFFT-PD-Swin(ours)	**95.98±0.26**	**93.98±0.43**

Apart from OA, the CMs of the proposed HFFT-PD-Poolformer are shown in Fig. 13 under different training ratios. It can be found from Fig. 13(a), 40 of 45 categories achieve an accuracy of over 90% when the training ratio is 20%. With a training ratio of 10%, there are 36 of 45 categories that maintain an accuracy of over 90% as shown in Fig. 13(b).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 13. CMs of HFFT-PD on the NWPU dataset with (a) 20% and (b) 10% training ratios.

TABLE VI

COMPARISON OF $F1$-SCORE AND SPECIFICITY WITH DIFFERENT MODELS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Architecture	F1-score (%)						Specificity (%)					
	UCM		AID		NWPU		UCM		AID		NWPU	
	80%	50%	50%	20%	20%	10%	80%	50%	50%	20%	20%	10%
VGG-VD-16 [51]	88.12	87.85	73.34	69.9	63.29	59.99	99.36	99.39	99.27	99.14	99.27	99.19
ResNet50 [52]	95.64	92.78	91.28	87.94	84.92	81.79	99.78	99.64	99.71	99.60	99.66	99.59
Vit-B [19]	97.74	96.79	95.38	93.39	92.10	89.43	99.89	99.84	99.85	99.78	99.82	99.76
PVT [28]	99.08	98.13	95.20	93.89	91.22	89.83	99.95	99.91	99.84	99.80	99.80	99.77
Poolformer [30]	98.36	96.64	95.12	92.74	92.27	89.92	99.92	99.83	99.85	99.76	99.82	99.77
Swin [29]	99.17	97.74	96.34	95.40	93.65	92.05	99.96	99.89	99.88	99.85	99.86	99.82
HFFT-PD-PVT (ours)	99.09	98.44	95.61	94.13	91.72	90.41	99.95	99.92	99.85	99.81	99.81	99.78
HFFT-PD-Poolformer (ours)	**99.45**	**98.44**	96.04	94.86	93.27	91.25	**99.97**	**99.92**	99.87	99.83	99.85	99.80
HFFT-PD-Swin (ours)	99.30	98.10	**96.90**	**95.57**	**94.43**	**92.12**	99.96	99.91	**99.90**	**99.85**	**99.87**	**99.82**

From the above results, our HFFT-PD still performs well despite the NWPU dataset being complex and large-scale.

2) Comparison of F1-Score and Specificity With Other Models: In order to comprehensively evaluate the performance of our method, the $F1$-score and Specificity are the other evaluation metrics to further evaluate the performance of the proposed model.

By comparing the $F1$-score and Specificity metrics with other models, the superiority of our proposed mode is further demonstrated. From Table VI, it can be observed that the obtained $F1$-score exhibits a consistent conclusion as OA from the experimental results, indicating that the performance of the Transformer model is superior to that of the CNN models, and our model outperforms their corresponding Transformer backbones, demonstrating that our model achieves the best performance. From the results of Specificity, it can be observed that most models demonstrate good predictive ability for negative samples, while our method still achieves the best results. This fully validates that the proposed model has better stability and generalization.

Based on the results of the above experiments in the three datasets, we can draw the following results.

1) The Transformer-based models obtained superior results compared to the CNN-based models on almost all datasets, which confirms the potential of the Transformer model in RS scene classification.
2) Compared with the original Transformer backbones, the proposed HFFT-PD model combined with the Transformer backbones shows better performance than the corresponding backbones on the scene classes with complex spatial features. This shows that the multilayer feature fusion method can improve the performance of RS scene classification.
3) Compared with the Tranformer-based state-of-the-art models, the proposed HFFT-PD-Swin model achieves the best classification performance in almost all cases. This shows that our proposed approach is not only superior to the CNN-based approaches but also shows excellent performance against the existing transformer-based approaches.

3) Comparison of Number of Parameters With Other Models: To evaluate the models, the total parameters of the model are compared. In order to balance the number of parameters and classification performance, the ViT-base, Poolformer-small, PVT-small, and Swin-tiny are selected to execute the experiments. As can be seen in Table VII, the VGG-VD-16 has the most parameters but the worst OA in all cases. Although ResNet50 has a small number of parameters, its OA is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
COMPARE PARAMETERS WITH OTHER MODELS ON THE AID DATASET UNDER 20% TRAINING RATIO

Methods	Parameters	OA (%)
VGG-VD-16 [51]	138.4 M	86.59
ResNet50 [52]	25.6 M	93.64
Vit-Base [19]	86.4 M	94.90
PVT-Small [28]	24.0 M	95.59
Poolformer-Small [30]	30.4 M	94.32
Swin-Tiny [29]	27.5 M	96.24
SCViT [27]	40.1 M	95.56
HFFT-PD+PVT-Small	25.3 M	96.04
HFFT-PD+Poolformer-Small	31.7 M	96.01
HFFT-PD+Swin-Tiny	29.3 M	**97.08**

consistently inferior to that of the Transformer-based methods. Compared to Swin-tiny and PVT-small, ViT-base specifically has a larger number of parameters, that is because it applies the "columnar" structure, while our method obtains a better performance with a tiny increase in parameters relative to the corresponding backbones. The above analysis illustrates that our method makes a trade-off between parameter complexity and performance. These results indicate that our proposed HFFT-PD is practical for the task of RS scene classification.

D. Verification of Generality

The proposed HFFT-PD is well combined with the hierarchical Transformer models (e.g., Poolformer, PVT, and Swin), and some experiments are conducted to verify the generality. To verify the generality of the HFFT-PD, Poolformer, PVT, and Swin are selected as the backbones which are three popular hierarchical Transformer architectures. The results of the experiments are shown in Fig. 14 for the backbone of Poolformer, Fig. 15 for the backbone of PVT, and Fig. 16 for the backbone of Swin. It can be seen that the HFFT-PD combined with any backbone outperforms the corresponding original backbones. Therefore, it can be concluded that the proposed HFFT-PD can easily combine with various hierarchical Transformer architectures, and expand the scope and impact of the application.

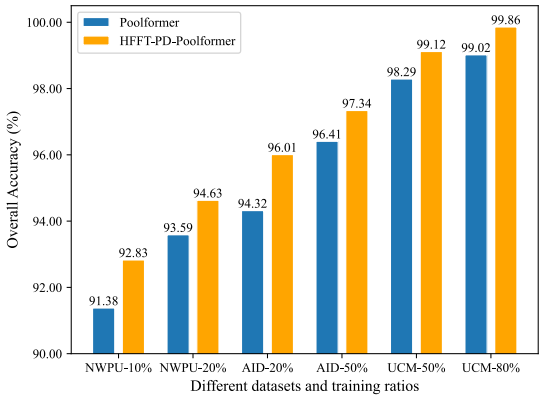


Fig. 14. Verification of generality on the backbone of Poolformer.

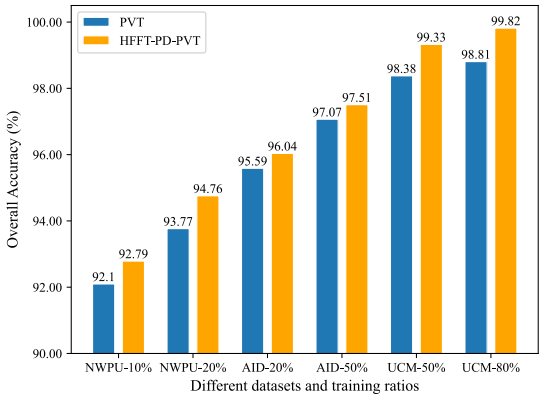


Fig. 15. Verification of generality on the backbone of PVT.

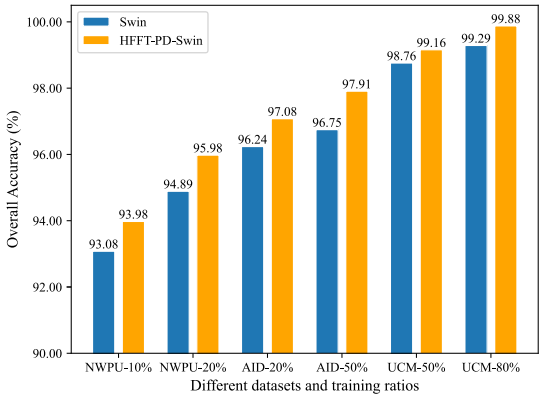


Fig. 16. Verification of generality on the backbone of Swin.

E. Discussions

1) Ablation Study: To verify the contributions of the different components in the proposed HFFT-PD, ablation experiments are conducted on three datasets. In this section, all ablation experiments are performed on the backbone of the Poolformer model.

a) Effects of different blocks: The proposed method consists of two blocks, including HTM and LACC. The HTM block aims at mining multilevel and multiscale information within the scene of RS images, while the LACC in the classification block is designed to focus on the critical features and eliminate redundant information among other features before the final classifier. The importance of each block is studied in this part. To inspect the significance of each module, the following models are constructed for comparison.

1) *Net-0:* Poolformer (without HTM block and without LACC module).
2) *Net-1:* Poolformer + LACC (without HTM module).
3) *Net-2:* Poolformer + HTM (without LACC module).
4) *Net-3:* Poolformer + HTM + LACC (our proposed).

i) Effects of LACC: First, the HTM block is fixed to analyze the contributions of the LACC module. A lot of compared experiments are conducted on three datasets under different training ratios. The results of the experiments are shown in Table VIII. *Net-3* and *Net-2* are with and without the LACC module, and it can be found from Table VIII, the result of *Net-3* are higher than *Net-2* on three datasets, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
VALIDATION OF DIFFERENT MODULES

Architecture	UCM		AID		NWPU	
	80%	50%	50%	20%	20%	10%
Net-0	99.02	98.29	96.41	94.32	93.59	91.38
Net-1	99.29	99.05	97.04	95.17	94.32	92.23
Net-2	99.26	98.81	96.62	95.06	94.21	91.79
Net-3 (ours)	**99.86**	**99.12**	**97.34**	**96.01**	**94.63**	**92.83**

TABLE IX
VALIDATION OF MPF MODULE AND FM MODULE IN THE HTM BLOCK

Architecture	UCM		AID		NWPU	
	80%	50%	50%	20%	20%	10%
Scheme-0	99.76	99.05	97.08	95.54	94.44	92.63
Scheme-1	99.52	99.06	97.28	95.76	94.33	92.67
Scheme-2 (ours)	**99.86**	**99.12**	**97.34**	**96.01**	**94.63**	**92.83**

While *Net-1* obtains higher OA than *Net-0* which is just a backbone. Therefore, it can be illustrated that the LACC module is useful for focusing on the crucial channels to boost classification performance.

ii) Effects of HTM: To verify the contribution of the HTM block, the LACC module is fixed. From the observation in Table VIII, the same conclusion can be made. Specifically, the contribution of the HTM block is positive to classification performance. The performance of *Net-3* outperforms *Net-1* on three datasets, respectively. While *Net-2* gets higher OA than *Net-0* which gets the worst result of OA. These results illustrate the importance of mining the multiscale information within the scene of RS images.

iii) Effects of HTM + LACC: From the above conclusion, it can easily be found that adding the HTM and LACC simultaneously achieves the best performance in all cases. *Net-1* and *Net-2* only add the LACC and the HTM, respectively, which both perform worse than *Net-3*. The result further illustrates the effectiveness of our proposed method which is beneficial to exploit the multiscale and complex contents within scenes of RS images.

From the above ablation experimental results, it can be seen that our proposed HFFT-PD exhibits superior classification performance. The reason is that our HFFT-PD is equipped with the ability to learn both multiscale information and enhanced key information, which enables the identification of complex hidden content and intrinsic relationships within RS scenes.

b) Effects of MPF and FM in the HTM: In this part, the contribution of the MPF module and FM module are investigated in the HTM block. As above mentioned in Section III-A, the MPF first merge the adjacent feature maps to diminish the semantic gaps of different level, while the FM mixes and aggregates these features to explore the correlations of different level features. Therefore, the LACC module is fixed first, then the investigation is conducted in the HTM block. For comparison, the scheme is constructed in the following.

1) *Scheme-0:* Without MPF.
2) *Scheme-1:* Without FM.
3) *Scheme-2:* HFFT-PD (ours).

TABLE X
COMPARISON WITH DIFFERENT UPSAMPLING STRATEGIES

Upsampling	UCM		AID		NWPU	
	80%	50%	50%	20%	20%	10%
Bilinear	99.52	98.86	97.32	95.38	94.14	92.50
Deconvolution	99.76	98.95	97.02	95.56	94.23	92.25
Pixelshuffle	99.52	98.76	97.12	95.80	94.41	92.43
Patch Dilating (ours)	**99.86**	**99.12**	**97.34**	**96.01**	**94.63**	**92.83**

i) Effects of MPF: To illustrate the contribution of the MPF, the FM is fixed, and the feature is from the last layer of the backbone. From Table IX, it can be found between *Scheme-2* and *Scheme-0*, the biggest and the smallest gaps are 0.46% and 0.07% on the AID datasets with 20% training ratio and UCM datasets with 50 % training ratio, respectively. Meanwhile, the results of *Scheme-2* are all better than *Scheme-0* on the corresponding datasets. Therefore, it verifies the module of MPF is beneficial to improve the classification performance.

ii) Effects of FM: The same strategy as above is applied fixing the MPF first to verify the effect of the FM. It can be seen from Table IX that the same conclusion can be made: *Scheme-1* performs worse than *Scheme-2* on all datasets. These results illustrate the contribution of the FM is positive for scene classification performance.

As can be seen from Table IX, the results are lower when either MPF or MF is missing, compared to inserting both MPF and FM modules. The result illustrates the effectiveness of the two modules.

2) Discussion of Different Upsampling Strategies: In this part, the effectiveness of the Patch Dilating is introduced which is used to upsample higher-level feature maps to align the spatial resolution of adjacent lower-level feature maps. To illustrate the superiority of the proposed upsampling strategy, the traditional upsampling methods, Deconvolution and Pixelshffle are compared with ours. From Table X, it can be found that our proposed Patch Dilating achieves the best results in all cases, especially on the AID dataset under 20% training ratio, there is an improvement of 0.63%. Deconvolution falls short compared to our method, as it requires more learnable parameters, which not only increases computational and storage costs but also could lead to overfitting. Although Pixleshuffle obtains near optimal results, our upsampling strategy outperforms it in all cases. This further illustrates that the patch-based upsampling strategy is more suitable for the Transformer structure than the pixel-based upsampling strategy. Therefore, it certifies our proposed strategy is effective for the Transformer architecture.

3) Discussion of Different Fusing Strategies: The MPF module is introduced to merge the multilevel feature maps between the adjacent Transformer block. To illustrate the advantage of our fusing strategy, the FPN [17] and Concatenate strategy are compared. For the fairness of the experiment, the MPF module is only replaced in our model, while all other modules remain unchanged. For the Concatenate strategy, we directly concatenate the obtained feature maps of each layer after the GAP operation. It can be found from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
COMPARISON WITH DIFFERENT FUSING STRATEGIES

Architecture	UCM		AID		NWPU	
	80%	50%	50%	20%	20%	10%
Concatenate	99.52	98.76	96.71	95.29	94.41	92.49
FPN [17]	99.76	98.57	96.66	95.41	94.36	92.35
MPF (ours)	**99.86**	**99.12**	**97.34**	**96.01**	**94.63**	**92.83**

TABLE XII
DISCUSSION OF DIFFERENT KERNEL SIZE

Kernel size	UCM		AID		NWPU	
	80%	50%	50%	20%	20%	10%
K=1	99.76	99.04	97.20	95.61	94.41	92.78
K=3	99.76	99.05	97.24	95.76	94.44	92.79
K=5	**99.86**	**99.12**	**97.34**	**96.01**	**94.63**	**92.83**
K=7	99.76	98.67	97.12	95.65	94.43	92.36

Table XI, the proposed MPF achieves the best results in all cases. Especially, our method is 0.63% and 0.72% higher than FPN on the AID datasets with training ratios of 50% and 20%, respectively. This is because our fusion strategy takes into account the spatial localization of image features and better meets the needs of Transformer-based feature extraction, which is due to the Transformer being based on patches.

4) Discussion of Different Kernel Size: The convolution kernel size (labeled as K) directly affects the effectiveness of the LACC module. Therefore, in this section, the impact of different K on the performance of the LACC module is validated. In the section, all experiments are performed on the backbone of the Poolform. In the experiment, the OA values are validated when $K = \{1, 3, 5, 7\}$, respectively. The specific results are shown in Table XII. From Table XII, it can be seen that when $K = 5$, the best OA was achieved in all datasets. Therefore, according to the results, the convolution kernel size chooses $K = 5$ in the LACC module.

V. CONCLUSION

In RS image scene classification, it is crucial to consider the combination of multiple target categories and the contextual relationships among these targets in order to achieve a comprehensive understanding of image scenes. Although Transformers are capable of capturing long-range contextual information, traditional Transformers almost disregard the intermediate features learned in intermediate layers of Transformer, which contain detailed spatial and multiscale information. In this article, a general model HFFT-PD is proposed to improve the classification performance of RS scenes. The model mainly consists of three modules, including the MPF module, the MF module, and the LACC module. The MPF module is introduced to merge the multilevel feature maps for the different Transformer blocks to narrow the semantic gaps between adjacent feature maps. Furthermore, to explore correlations of muti-level features, the FM module is added following the MPF module to aggregate and mix the obtained features from the MPF module. By fusing the multilayer features, a large number of features are obtained, which contain masses of redundant information that affect the classification effect. Therefore, the LACC module is designed, which adaptively weigh the features to enhance the important information and eliminate the redundant information. Experimental results on three widely used datasets show that our HFFT-PD model compares positively for RS scene classification tasks.

Meanwhile, the HFFT-PD is a general multilevel feature enhancement method that can be easily connected to different hierarchical Transformer models. Therefore, the proposed model can be expanded to more different fields, not only for RS images but also for nature images. The future work will be to investigate the application of our method to other fields.

REFERENCES

[1] Q. Zhu, Y. Zhong, L. Zhang, and D. Li, "Adaptive deep sparse semantic modeling framework for high spatial resolution image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6180–6195, Oct. 2018.

[2] X. Zheng, Y. Yuan, and X. Lu, "A deep scene representation for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4799–4809, Jul. 2019.

[3] Y. Yi, Z. Zhang, W. Zhang, H. Jia, and J. Zhang, "Landslide susceptibility mapping using multiscale sampling strategy and convolutional neural network: A case study in Jiuzhaigou region," *Catena*, vol. 195, Dec. 2020, Art. no. 104851.

[4] M. Fingas and C. Brown, "A review of oil spill remote sensing," *Sensors*, vol. 18, no. 2, p. 91, Dec. 2017.

[5] R. Avtar et al., "Exploring renewable energy resources using remote sensing and GIS—A review," *Resources*, vol. 8, no. 3, p. 149, Aug. 2019.

[6] K. C. Tiwari, M. K. Arora, and D. Singh, "An assessment of independent component analysis for detection of military targets from hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 13, no. 5, pp. 730–740, Oct. 2011.

[7] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[8] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[9] R. Minetto, M. Pamplona Segundo, and S. Sarkar, "Hydra: An ensemble of convolutional neural networks for geospatial land classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6530–6541, Sep. 2019.

[10] X. Wang, S. Wang, C. Ning, and H. Zhou, "Enhanced feature pyramid network with deep semantic embedding for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7918–7932, Sep. 2021.

[11] X. Tang, Q. Ma, X. Zhang, F. Liu, J. Ma, and L. Jiao, "Attention consistent network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2030–2045, 2021.

[12] D. Wang and J. Lan, "A deformable convolutional neural network with spatial-channel attention for remote sensing scene classification," *Remote Sens.*, vol. 13, no. 24, p. 5076, Dec. 2021.

[13] C. Xu, G. Zhu, and J. Shu, "A lightweight and robust lie group-convolutional neural networks joint representation for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501415.

[14] X. Chen, M. Ma, Y. Li, and W. Cheng, "Fusing deep features by kernel collaborative representation for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 12429–12439, 2021.

[15] K. Xu, H. Huang, Y. Li, and G. Shi, "Multilayer feature fusion network for scene classification in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 11, pp. 1894–1898, Nov. 2020.

[16] T. Gao, Z. Liu, J. Zhang, G. Wu, and T. Chen, "A task-balanced multiscale adaptive fusion network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613515.

[17] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.

[18] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
[19] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[20] H. Touvron et al., "ResMLP: Feedforward networks for image classification with data-efficient training," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 5314–5321, Apr. 2023.
[21] L. Yuan et al., "Tokens-to-Token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 538–547.
[22] J. Li et al., "Next-ViT: Next generation vision transformer for efficient deployment in realistic industrial scenarios," 2022, *arXiv:2207.05501*.
[23] I. Tolstikhin et al., "MLP-mixer: An all-MLP architecture for vision," in *Proc. 35th Conf. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 24261–24272.
[24] Y. Bazi, L. Bashmal, M. M. A. Rahhal, R. A. Dayil, and N. A. Ajlan, "Vision transformers for remote sensing image classification," *Remote Sens.*, vol. 13, no. 3, p. 516, Feb. 2021.
[25] J. Zhang, H. Zhao, and J. Li, "TRS: Transformers for remote sensing scene classification," *Remote Sens.*, vol. 13, no. 20, p. 4143, Oct. 2021.
[26] W. Li, D. Cao, Y. Peng, and C. Yang, "MSNet: A multi-stream fusion network for remote sensing spatiotemporal fusion based on transformer and convolution," *Remote Sens.*, vol. 13, no. 18, p. 3724, Sep. 2021.
[27] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409512.
[28] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 548–558.
[29] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
[30] W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10809–10819.
[31] S. D'Ascoli et al., "ConViT: Improving vision transformers with soft convolutional inductive biases," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 2286–2296.
[32] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.
[33] L. Ye, M. Rochan, Z. Liu, and Y. Wang, "Cross-modal self-attention network for referring image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10494–10503.
[34] K. Wu, H. Peng, M. Chen, J. Fu, and H. Chao, "Rethinking and improving relative position encoding for vision transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10013–10021.
[35] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10347–10357.
[36] J. Guo et al., "CMT: Convolutional neural networks meet vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12165–12175.
[37] M.-H. Guo, Z.-N. Liu, T.-J. Mu, and S.-M. Hu, "Beyond self-attention: External attention using two linear layers for visual tasks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5436–5447, May 2023.
[38] H. Wu et al., "CvT: Introducing convolutions to vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 22–31.
[39] Q. Hou, Z. Jiang, L. Yuan, M.-M. Cheng, S. Yan, and J. Feng, "Vision permutator: A permutable MLP-like architecture for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 1328–1334, Jan. 2023.
[40] M. Chen et al., "Generative pretraining from pixels," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1691–1703.
[41] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
[42] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[43] K. Xu, P. Deng, and H. Huang, "Vision transformer: An excellent teacher for guiding small networks in remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618715.
[44] Y. Zhao, J. Liu, J. Yang, and Z. Wu, "EMSCNet: Efficient multisample contrastive network for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605814.
[45] M. Bi, M. Wang, Z. Li, and D. Hong, "Vision transformer with contrastive learning for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 738–749, 2023.
[46] P. Deng, K. Xu, and H. Huang, "When CNNs meet vision transformer: A joint framework for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[47] X. Tang, M. Li, J. Ma, X. Zhang, F. Liu, and L. Jiao, "EMTCAL: Efficient multiscale transformer and cross-level attention learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626915.
[48] G. Wang, N. Zhang, W. Liu, H. Chen, and Y. Xie, "MFST: A multi-level fusion network for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[49] J. Wang, K. Chen, R. Xu, Z. Liu, C. C. Loy, and D. Lin, "CARAFE: Content-aware reassembly of features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3007–3016.
[50] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.
[51] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[52] Z. Zhao, J. Li, Z. Luo, J. Li, and C. Chen, "Remote sensing image scene classification based on an enhanced attention module," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 11, pp. 1926–1930, Nov. 2021.
[53] R. M. Anwer, F. S. Khan, J. van de Weijer, M. Molinier, and J. Laaksonen, "Binary patterns encoded convolutional neural networks for texture recognition and remote sensing scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 74–85, Apr. 2018.
[54] Q. Bi, K. Qin, H. Zhang, J. Xie, Z. Li, and K. Xu, "APDC-Net: Attention pooling-based convolutional network for aerial scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 9, pp. 1603–1607, Sep. 2020.
[55] W. Zhang, P. Tang, and L. Zhao, "Remote sensing image scene classification using CNN-CapsNet," *Remote Sens.*, vol. 11, no. 5, p. 494, Feb. 2019.
[56] C. Shi, T. Wang, and L. Wang, "Branch feature fusion convolution network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5194–5210, 2020.
[57] K. Xu, H. Huang, P. Deng, and Y. Li, "Deep feature aggregation framework driven by graph convolutional network for scene classification in remote sensing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 10, pp. 5751–5765, Oct. 2022.
[58] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[59] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2019.
[60] R. Cao, L. Fang, T. Lu, and N. He, "Self-attention-based deep feature fusion for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 43–47, Jan. 2021.



Xiaoning Chen received the B.S. degree in electronic information science and technology and the M.S. degree in communication and information systems from Northwest University, Xi'an, China, in 2004 and 2011, respectively. She is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an.

Her research interests include computer vision and remote sensing image analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.



Mingyang Ma (Student Member, IEEE) received the B.S. degree in communication engineering and the Ph.D. degree in communication and information systems from Northwestern Polytechnical University, Xi'an, China, in 2015 and 2021, respectively.

He is currently an Assistant Professor with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an. His main research interests include image processing and video summarization.



Yong Li received the B.S. degree in electronic engineering and the M.S. and Ph.D. degrees in circuits and systems from Northwestern Polytechnical University, Xi'an, China, in 1983, 1988, and 2005, respectively.

In 1983, he joined the School of Electronic Information, Northwestern Polytechnical University, where he has been a Professor since 2002. His research interests include digital signal processing and radar signal processing.



Shaohui Mei (Senior Member, IEEE) received the B.S. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2011, respectively.

He was a Visiting Student with the University of Sydney, Sydney, Australia, from October 2007 to October 2008. He is currently an Associate Professor with the School of Electronics and Information, Northwestern Polytechnical University. His research interests include hyperspectral remote sensing image processing and applications, intelligent signal and information acquisition and processing, video processing, and pattern recognition.

Dr. Mei was a recipient of the Excellent Doctoral Dissertation Award of Shaanxi Province in 2014 and the Best Paper Award of IEEE Intelligent Signal Processing and Communication Systems (ISPACS) in 2017. He has been serving as a reviewer for more than 20 international famous academic journals. He was awarded as the Best Reviewer of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2019. He has served as the Registration Chair of IEEE China Summit and International Conference on Signal and Information Processing (ChinaSIP) in 2014.



Zonghao Han received the B.S. degree in communication engineering from Ningxia University, Yinchuan, China, in 2020. He is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His major research interests include hyperspectral remote sensing image processing and deep learning.



Jian Zhao received the B.S. degree in applied electronics from the Xi'an Institute of Technology, Xi'an, China, in 1995, and the M.S. degree in signal and information processing and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, in 1999 and 2003, respectively.

He is currently a Professor with the School of Information Science and Technology, Northwest University, Xi'an. His research interests include human–machine interaction and artificial intelligence.



Wei Cheng received the B.S. degree in electronic information engineering and the M.S. and Ph.D. degrees in communication and information system from Northwestern Polytechnical University, Xi'an, China, in 2003, 2006, and 2011, respectively.

He is currently working as an Associate Professor with the School of Electronic Information, Northwestern Polytechnical University. His research interests include wireless sensor networks, ad hoc networks, machine learning, and radar signal processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:33:28 UTC from IEEE Xplore. Restrictions apply.